

Applied Optimization

Exercise 1 - Convex Sets

September 18, 2025

What to hand in

A .zip compressed file renamed to `Exercise $n$ -Group.zip` where n is the number of the current exercise sheet. It should contain:

- Hand in **only** the files you changed (headers and source). It is up to you to make sure that all files that you have changed are in the zip.
- A `readme.txt` file containing a description on how you solved each exercise (use the same numbers and titles) and the encountered problems. Indicate what fraction of the total workload each project member contributed.
Failure to do so will result in you losing a point.
- Other files that are required by your `readme.txt` file. For example, if you mention some screenshot images in `readme.txt`, these images need to be submitted too.
- For the theory exercise, put `TheoryExercise.pdf` with your solutions in the same .zip you submit for the code.
- Submit your solutions to ILIAS before the submission deadline. Late submissions will receive 0 points!
- **Important:** Your submission MUST compile without any issue in order to obtain a passing grade.

Convex Sets (1 pt)

Example sets

Sketch the following sets in $\mathbb{R}^2$

1. $\text{span} \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -0.5 \\ -0.5 \end{pmatrix} \right\}$
2. $\text{span} \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0.5 \\ -0.5 \end{pmatrix} \right\}$
3. $\text{aff} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$
4. $\text{conv} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ -2 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 1 \end{pmatrix}, \begin{pmatrix} -2 \\ -2 \end{pmatrix} \right\}$

Convexity

Let $C \subset \mathbb{R}^n$ be a convex set, with $x_1, \dots, x_k \in C$, and let $\theta_1, \dots, \theta_k \in \mathbb{R}$ satisfy $\theta_i \geq 0, \theta_1 + \dots + \theta_k = 1$. Show that $\theta_1 x_1 + \dots + \theta_k x_k \in C$. (The definition of convexity is that this holds for $k = 2$; you must show it for arbitrary k .) Hint. Use induction on k .

Linear Equations

Show that the solution set of linear equations $\{x | Ax = b\}$ with $x \in \mathbb{R}^n$, $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$ is an affine set.

Linear Inequalities

1. Show that the solution set of linear inequalities $\{x | Ax \preceq b, Cx = d\}$ with $x \in \mathbb{R}^n$, $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$, $C \in \mathbb{R}^{k \times n}$ and $d \in \mathbb{R}^k$ is a convex set. Here $\preceq$ means componentwise less or equal.

2. Is it an affine set?

Voronoi description of halfspace

Let a and b be distinct points in $\mathbb{R}^n$. Show that the set of all points that are closer (in Euclidean norm) to a than b , i.e., $\{x \mid \|x - a\|^2 \leq \|x - b\|^2\}$, is a halfspace. Describe it explicitly as an inequality of the form $c^T x \leq d$. Draw a picture.

Convex Illumination Problem (1 pt)

Show that the solution $p^* = (p_1^*, p_2^*, \dots, p_n^*)^T \in \mathbb{R}^n$ of the non-convex illumination problem from the lecture

$$\begin{array}{ll} \text{minimize} & \max_{k=1 \dots m} |\log I_k - \log I_{des}| \\ \text{subject to} & 0 \leq p_j \leq p_{max}, \quad j = 1 \dots n \end{array}$$

with $I_k = \sum_{j=1}^n a_{kj} p_j$ for geometric constants $a_{kj} \in \mathbb{R}$, a constant desired illumination $I_{des} \in \mathbb{R}$ and an upper bound $p_{max} \in \mathbb{R}$ on the lamp power, is identical to the solution of the following equivalent (convex) problem

$$\begin{array}{ll} \text{minimize} & \max_{k=1 \dots m} h(I_k / I_{des}) \\ \text{subject to} & 0 \leq p_j \leq p_{max}, \quad j = 1 \dots n \end{array}$$

with $h(u) = \max\{u, 1/u\}$.